INCH-POUND
MIL-PRF-32664
29 September 2020

PERFORMANCE SPECIFICATION

DECK PANELS, FALSE, COMPOSITE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers composite panels for false decking used in naval shipboard interior applications that are installed no greater than 15 inches above the true deck. This also includes the false deck tile configuration that is used in flexible infrastructure installations.

1.2 Classification (see 6.2).

1.2.1 Types. The types of false deck panels are as follows:

a. Type I – False deck panels manufactured using any combustible material.

b. Type II – False deck panels manufactured using only non-combustible materials in accordance with ISO 1182.

1.2.2 Classes. The classes of false deck panels are as follows:

a. Class 1 – False deck panels that are manufactured with a wear surface that is in accordance with the requirements of MIL-STD-1623.

b. Class 2 – False deck panels that are manufactured with a wear surface that is not in accordance with the requirements of MIL-STD-1623.

1.2.3 Forms. False deck panels are classified by form based on whether the panels are finished or unfinished.

a. Form 1 – False deck panels that are unfinished. The false deck panels' maximum size does not exceed 48 inches wide by 100 inches long (see 6.2). These panels are intended to be cut to size by the installing activity as part of the shipboard installation.

b. Form 2 – False deck panels that are finished. The false deck panels' size is a standard 24 by 10⅜ inches in areal dimension with edges that are finished as shown on figure 1. These false deck panels are intended for use in flexible infrastructure applications without modification by the installing activity as part of the shipboard installation into the flexible infrastructure track system.

1.2.4 Groups. The false deck panels are classified by group based on whether the panels are to be installed in electrical spaces or in non-electrical spaces.

a. Group 1 – False deck panels manufactured with materials that provide safety protection from accidental exposure to electrical potentials in electrical spaces.

b. Group 2 – False deck panels that are considered general purpose for installation in non-electrical spaces.

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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1.2.5 Areal weight.

a. Areal weight 1 – For lightest deck panels where weight is most critical.

b. Areal weight 2 – For applications where weight is less critical.

c. Areal weight 3 – For applications where weight is not critical.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-901	-	Shock Test, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for
MIL-DTL-15562	-	Matting or Sheet, Floor Covering Insulating for High Voltage Application
MIL-PRF-32170	-	Deck Tiles, Wear-Resistant
DOD-A-82720	-	Adhesive, Modified-Epoxy, Flexible, Two-Part (Metric)

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1623	-	Fire Performance Requirements and Approved Specifications for Interior Finish Materials and Furnishings (Naval Shipboard Use)
MIL-STD-1916	-	DOD Preferred Methods for Acceptance of Products

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) DRAWINGS

802-8697457	-	CVN68 Class Design Guidance for Flexible Infrastructure (FI) Deck Tile and Track Cover Fabrication

(Copies of this document are available from the applicable repositories listed in S0005-AE-PRO-010/EDM, which can be obtained online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/. Copies of this document may also be obtained from the Naval Ships Engineering Drawing Repository (NSEDR) online at https://199.208.213.105/webjedmics/index.jsp. To request an NSEDR account for drawing access, send an email to NNSY_JEDMICS_NSEDR_HELP_DESK@navy.mil.)

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NAVSEA PUBLICATIONS

T9070-AK-DPC-010/078-1	-	Composite Materials, Surface Ships, Topside Structural and Other Topside Applications – Fire Performance Requirements
T9070-AL-DPC-020/077-2	-	NAVSEA Hazardous Material Avoidance Process

(Copies of these documents are available online via TDMIS at https://mercury.tdmis.navy.mil/. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. These documents are available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

NATIONAL INSTITUTE FOR OCCUPATIONAL SAFETY AND HEALTH (NIOSH)

Immediately Dangerous to Life or Health (IDLH) Values	-	Table of IDLH Values

(Copies of this document are available online at www.cdc.gov/niosh/idlh/intridl4.html.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AEROSPACE INDUSTRIES ASSOCIATION (AIA)

NAS 411-1	-	Hazardous Material Target List

(Copies of this document are available online at www.aia-aerospace.org.

ASTM INTERNATIONAL

ASTM C297/C297M	-	Standard Test Method for Flatwise Tensile Strength of Sandwich Constructions
ASTM C364/C364M	-	Standard Test Method for Edgewise Compressive Strength of Sandwich Constructions
ASTM C365/C365M	-	Standard Test Method for Flatwise Compressive Properties of Sandwich Cores
ASTM C393/C393M	-	Standard Test Method for Core Shear Properties of Sandwich Constructions by Beam Flexure
ASTM D523	-	Standard Test Method for Specular Gloss
ASTM D570	-	Standard Test Method for Water Absorption of Plastics
ASTM D2047	-	Standard Test Method for Static Coefficient of Friction of Polish-Coated Flooring Surfaces as Measured by the James Machine
ASTM D2244	-	Standard Practice for Calculation of Color Tolerances and Color Differences from Instrumentally Measured Color Coordinates
ASTM D4060	-	Standard Test Method for Abrasion Resistance of Organic Coatings by the Taber Abraser
ASTM D7249/D7249M	-	Standard Test Method for Facesheet Properties of Sandwich Constructions by Long Beam Flexure
ASTM D7766/D7766M	-	Standard Practice for Damage Resistance Testing for Sandwich Constructions

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ASTM E162	-	Standard Test Method for Surface Flammability of Materials Using a Radiant Heat Energy Source
ASTM E662	-	Standard Test Method for Specific Optical Density of Smoke Generated by Solid Materials
ASTM E800	-	Standard Guide for Measurement of Gases Present or Generated During Fires
ASTM F1514	-	Standard Test Method for Measuring Heat Stability of Resilient Flooring by Color Change
ASTM F1515	-	Standard Test Method for Measuring Light Stability of Resilient Flooring by Color Change
ASTM F2055	-	Standard Test Method for Size and Squareness of Resilient Floor Tile by Dial Gage Method

(Copies of these documents are available online at www.astm.org.)

CEILINGS & INTERIOR SYSTEMS CONSTRUCTION ASSOCIATION (CISCA)

CISCA 109 - Recommended Test Procedures for Access Floors

(Copies of this document are available online at www.cisca.org.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 1182 - Reaction to Fire Tests for Products – Non-combustibility Test

ISO/IEC 17025 - General Requirements for the Competence of Testing and Calibration Laboratories

(Copies of these documents are available online at www.iso.org.)

SAE INTERNATIONAL

SAE AMS-STD-595 - Colors Used in Government Procurement

(Copies of this document are available online at www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

3.2 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

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3.3 Physical characteristics.

3.3.1 Workmanship. Materials shall be free from the following defects, which adversely affect performance or serviceability of the finished product (see 4.5.1.1):

a. Cracking
b. Scratches
c. Pits
d. Embedded foreign matter
e. Evidence of delamination
f. Broken edges
g. Voids
h. Non-uniform edges

3.3.2 Size. The false deck panels' size shall be as specified (see 6.2). The size of the panel shall not vary greater than $^{1}/_{64}$ inch per foot. The size is dependent on the form classification of the false deck panel (see 4.5.1.2).

a. Form 1. The maximum false deck panels' size that shall be available is 48 inches wide by 100 inches long.

b. Form 2. The false deck panels' size shall be a standard 24 by 10⅜ inches in areal dimension and shall have edges that are finished as shown on figure 1.

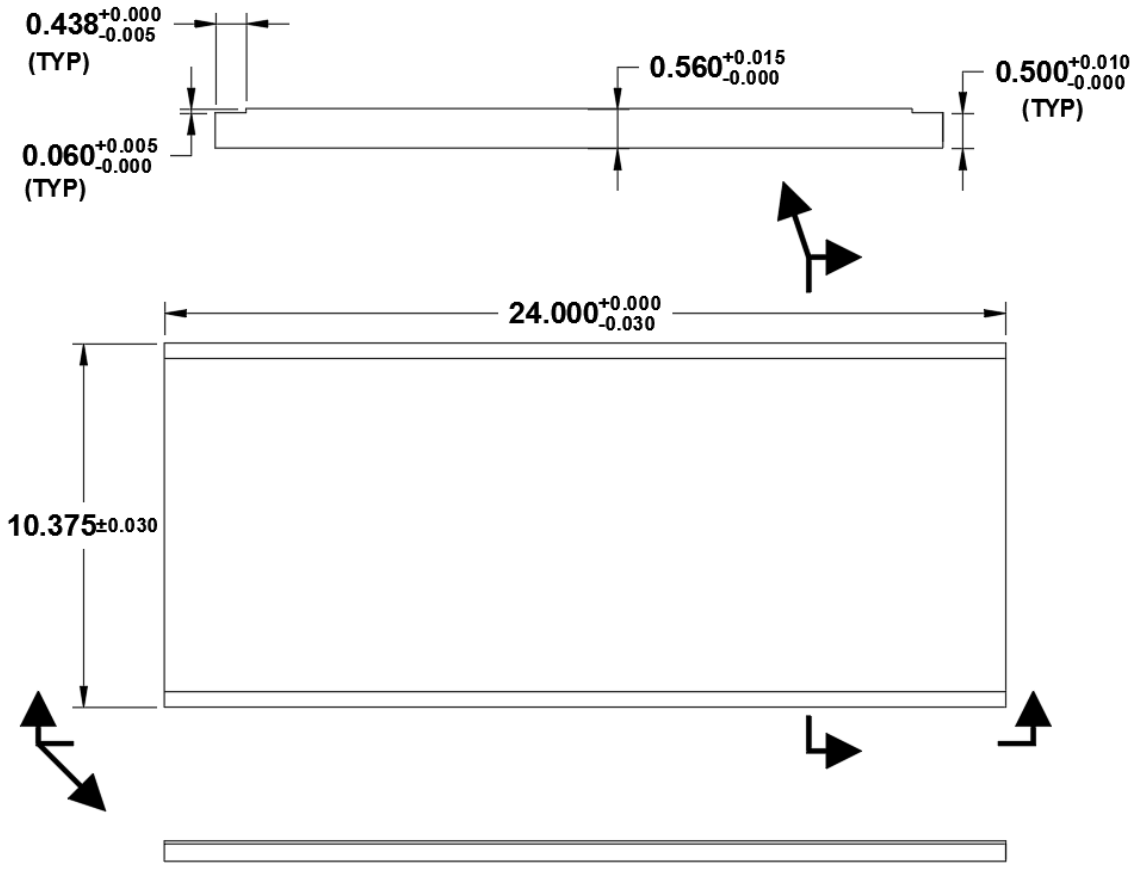


FIGURE 1. Flexible infrastructure deck tile design features.

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3.3.3 Thickness. The false deck panels shall be 0.560 inch+0.015 inch, -0.00 inch thick (see 4.5.1.3 and 6.2).

3.3.4 Symmetry. The panels shall have a horizontal symmetry through the center of the panel. Non-symmetric panels will be considered if the vendor can prove that the non-symmetric nature of the panels does not trap residual stress within the sandwich panel that might cause the panel to warp after installation in a shipboard environment over its expected life (see 6.2). An example of a non-symmetric panel would be one that has the wear surface only installed on one side of the panel (see 4.5.1.4).

3.3.5 Flatness. The panels shall have a maximum linear flatness of 0.016 inch per foot (see 4.5.1.5) with no open discontinuities.

3.3.6 Squareness. The perpendicularity of the false deck panels' edges shall not exceed 0.010 inch (see 4.5.1.6).

3.3.7 Areal weight. The false deck panel's areal weight shall be as specified (see 6.2). The areal weight is dependent on the areal weight classification of the false deck panel. While ranges of weight are given, in many cases, it is preferable for the panel to be as light as possible within the range given.

a. Areal weight 1. The false deck panel shall weigh less than or equal to 1.84 lb/ft^2 (see 4.5.1.7 and 6.2).

b. Areal weight 2. The false deck panel shall weigh greater than 1.84 lb/ft^2 and less than or equal to 2.5 lb/ft^2 (see 4.5.1.7 and 6.2).

c. Areal weight 3. The false deck panel shall weigh greater than 2.5 lb/ft^2 and less than or equal to 4.1 lb/ft^2 (see 4.5.1.7 and 6.2).

3.3.8 Color. The false deck panels' color shall be as specified (see 6.2) and maintain repeatability from one procurement to the next. The three most common colors required will be the basic or background (if mottled or marbleized pattern) color that shall be in accordance with semi-gloss blue (number 25177), semi-gloss gray (number 26231), or semi-gloss green (number 24260) of SAE AMS-STD-595 (see 4.5.1.8). If a sample pattern is required to be provided for examination (see 6.2), a solid, marbleized or mottled pattern shall match the sample pattern as approved by the procuring activity.

3.4 Electrical performance.

a. Group 1 false deck panels shall meet the following electrical performance requirements. Group 1 false deck panels shall resist the free flow of electrical current between the top and bottom surfaces and shall not puncture, become appreciably warm at any spot, or show any other sign of weakness when tested as specified.

b. Group 2 false deck panels are not required to meet the electrical performance requirements.

3.4.1 Dielectric strength. Group 1 false deck panels shall have a minimum dielectric strength of 30,000 volts alternating current (see 4.5.2.1).

3.4.2 Voltage. Group 1 false deck panels shall not puncture, become appreciably warm at any spot, or show any other signs of weakness when tested in accordance with MIL-DTL-15562 (see 4.5.2.2).

3.5 Combustibility/fire performance.

a. Type II false deck panels shall not contain combustible materials (see 4.5.4). Type II false deck panels are not required to meet fire performance requirements.

b. Type I false deck panels shall meet the following fire performance requirements (see 4.5.3). Each combustible layer shall separately meet the requirements of 3.5.1 through 3.5.3.

c. If the false deck panels are considered a class 1 material, the wear surface layer does not need to be tested to meet the requirements of 3.5.

d. If the false deck panels are considered a group 1 material that includes an electrical insulating sheet layer that meets all the requirements of MIL-DTL-15562, the electrical insulating sheet layer does not need be tested to meet the requirements of 3.5.

3.5.1 Flame spread index. False deck panels shall have a maximum flame spread index of 25 (see 4.5.3.2) with no evidence of melting, dripping, or flaming droplets (see 6.8).

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3.5.2 Smoke optical density. False deck panels shall have a maximum smoke optical density of 200 with no evidence of melting, dripping, or flaming droplets (see 4.5.3.3 and 6.8).

3.5.3 Fire gas toxicity. All false deck panels shall meet the fire gas toxicity requirements in accordance with T9070-AK-DPC-010/078-1, in both flaming and non-flaming modes, except the following limits shall be used (see 4.5.3.4 and 6.8):

a. Carbon monoxide (CO): 600 parts per million (ppm) (maximum).

b. Hydrogen chloride (HCL): 30 ppm (maximum).

c. Hydrogen cyanide (HCN): 30 ppm (maximum).

3.5.3.1 Fire gas immediately dangerous to life and health (IDLH) index. The IDLH Index (I_{IDLH}) of false deck panels shall be less than 1 using the following equation:

$$\text{Where } I_{IDLH} = \frac{C_{CO}}{IDLH_{CO}} + \frac{C_{HCL}}{IDLH_{HCL}} + \frac{C_{HCN}}{IDLH_{HCN}} + \frac{C_X}{IDLH_X}$$

Where $IDLH_{CO}$ = 1200 ppm, $IDLH_{HCL}$ = 50 ppm, and $IDLH_{HCN}$ = 50 ppm

3.5.3.2 Additional toxic gases. If candidate materials have a potential for significant concentrations of other toxic gases (C_x in the above formula), those gases shall be included in the fire gas IDLH index. $IDLH_x$ shall be in accordance with the IDLH concentration of fire gases of Immediately Dangerous to Life or Health (IDLH) Values.

3.6 Environmental durability. The false deck panels shall be capable of withstanding typical shipboard environmental conditions as represented by the requirements in 3.6 through 3.7.

3.6.1 Water resistance. The false deck panels shall be water resistant and shall show no signs of blistering, wrinkling, delamination, bulging, or discoloration when exposed to water. This shall include exposure to both the top surface and the sides of the panels (see 4.5.5.1).

3.6.2 Moisture absorption. The false deck panels shall resist warping, expansion, and contraction due to humid conditions and shall show no signs of blistering, wrinkling, delamination, or discoloration when exposed to water. This shall include exposure to both the top surface and the sides of the panels (see 4.5.5.2).

3.6.3 Resistance to cleaning agents. The false deck panels shall show no signs of blistering, wrinkling, discoloration, change in gloss, glistening, softening, swelling, loss of adhesion, or distinguishable difference when exposed to shipboard cleaning materials when using fluid wetted sponges. This shall include exposure to both the top surface and the sides of the panels. After 2 hours of air drying, the portion of the panel that was exposed to the fluid wetted sponges shall be indistinguishable with regard to hardness, color, and gloss from the unexposed areas of the false deck panel (see 4.5.5.3).

3.6.4 Heat and light resistance. The false deck panels shall resist discoloration and damage due to heat and light (see 4.5.5.4). The false deck panels shall have a measured change in color not greater than 5 ΔE (color difference) and shall show no appreciable delamination or signs of checking, cracking, or any other deterioration on their exposed surface (see 4.5.5.4).

3.6.5 Light reflectance. The outer surface of the false deck panels shall have a gloss rating of 25 through 60 (see 4.5.5.5).

3.6.6 Impact resistance. The false deck panels shall resist impact damage and shall show no signs of chipping, cracking, splintering, blistering, or delamination (see 4.5.5.6).

3.6.7 Shock resistance. The false deck panels shall resist damage or failure due to shock (see 4.5.5.7) as follows:

a. Form 1 false deck panels shall meet the grade B, class I shock requirements of MIL-DTL-901 (see 4.5.5.7).

b. Form 2 false deck panels shall meet the grade A, class I, type B shock requirements of MIL-DTL-901 (see 4.5.5.7).

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3.6.8 Slip resistance. The false deck panels shall have a static coefficient of friction of greater than or equal to 0.70 (see 4.5.5.8).

3.6.9 Wear resistance. The false deck panels shall resist wear and shall not show wear characteristics exceeding 0.001 ounce (30 milligrams) loss due to abrasion when tested in accordance with ASTM D4060 (see 4.5.5.9).

3.6.10 Rolling load resistance. False deck panels shall resist exposure to chair and equipment cart rolling wheels and shall exhibit no audible or visual evidence of damage to the core, facing, wear surface, or permanent set when exposed to a rolling load of 300 pounds (see 4.5.5.10).

3.7 Mechanical performance. The false deck panels shall be capable of withstanding normal operating conditions within a shipboard environment. These loads are primarily expected to be from foot traffic, equipment movement, and furniture.

3.7.1 Short beam shear strength. The false deck panels shall have a minimum average short beam shear strength of 270 pounds per square inch (psi) (see 4.5.6.1).

3.7.2 Long beam sandwich flexure load. The false deck panels shall have a minimum average ultimate maximum long beam sandwich flexure force of 225 pounds (see 4.5.6.2).

3.7.3 Sandwich compression strength. The false deck panels shall have a minimum average sandwich compression strength of 900 psi (see 4.5.6.3).

3.7.4 Edgewise compression strength. The false deck panels shall have a minimum average edgewise compression strength of 2,230 psi (see 4.5.6.4).

3.7.5 Flatwise tensile strength. The false deck panels shall have a minimum average flatwise tensile strength of 200 psi (see 4.5.6.5).

3.7.6 Maximum panel deflection. The maximum deflection requirement under an applied load shall be as follows:

a. Form 1 (unfinished) false deck panels' maximum deflection under a 300-pound load applied over a 10-inch by 10-inch area in the center of a 24-inch by 48-inch panel which is supported at a width of 0.5 inch along all edges shall be 0.16 inch while supporting the load (see 4.5.6.6).

b. Form 2 (finished) false deck panels' maximum deflection under a 300-pound load applied uniformly over a 4-inch by 10-inch area in the center of a 10.375-inch by 24-inch tile which is supported at a width of 1 inch along the 24-inch edges of the tile shall be of 0.20 inches while supporting the load. The 10-inch loading direction shall be aligned with the 24-inch side of the panel (see 4.5.6.6).

3.7.7 Panel load performance. The panel loaded according to the appropriate method in 3.7.6 shall exhibit no visual signs of splitting, cracking, or delamination. Form 1 (unfinished) false deck panels shall be loaded to 1,000 pounds. Form 2 (finished) false deck panels shall be loaded to 300 pounds (see 4.5.6.7).

3.7.7.1 Permanent deflection. The false deck panels shall resist permanent deflection after removal of the applied load (see 4.5.6.7).

3.8 Machinability. The false deck panels shall be modifiable using standard hand tools such as a hand-held circular saw or power drill. Modifications may include cutting the false deck panels to a smaller length in order to fit into a space or cutting a hole in a panel to allow for equipment cables. The false deck panels shall show no signs of chipping, cracking, splintering, blistering, or delamination due to standard machining operations (see 4.5.7).

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3.9 Interface capability. The interface capability requirement of the deck panels shall be as follows:

a. Form 1. The unfinished false deck panels shall be capable of being attached to the supporting framework using polyamide plastic (nylon) self-starting screws, ¼-inch FH 20 UNC-2A, 1½ inches long (or equivalent). Nylon ferrules, 5/16-inch inside diameter, Shur-Lok Company (or equivalent), SL 5229, may be installed (if necessary) in the panels along the edges located on 12-inch centers (maximum). The ferrules shall be molded to accommodate screws to a firm and flush fit and shall be secured in the panels using an epoxy adhesive in accordance with DOD-A-82720. The ferrule edges shall be ¼-inch ±0.030 from any edge (see 4.5.8).

b. Form 2. The finished false deck panels shall be capable of being placed on the flexible infrastructure deck track and held in place with the track cover as shown on figure 2. More detailed information regarding the shipboard installation details can be found on 802-8697457.

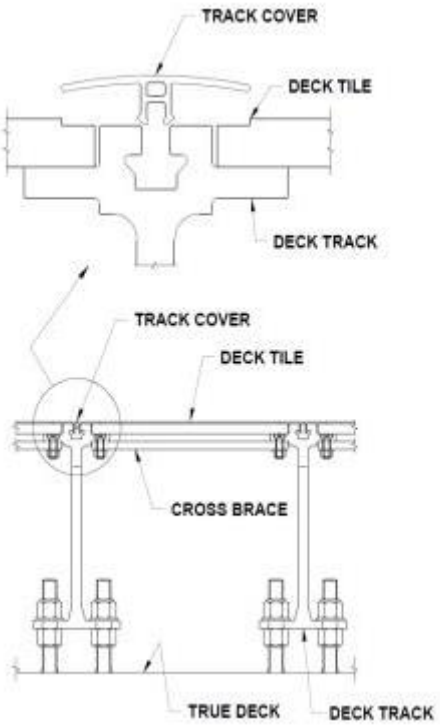


FIGURE 2. Flexible infrastructure (FI) false deck tile, track and cover interface.

3.10 Toxicity and prohibited materials.

3.10.1 Toxicity. When evaluated in accordance with 4.5.1.1, the false deck panels shall pose no serious or high risk to the health of personnel or the environment when used for its intended purpose (see 4.5.9 and 6.7).

3.10.2 Prohibited materials. The false deck panels shall not contain any chemicals categorized as “prohibited” in accordance with T9070-AL-DPC-020/077-2 for NAVSEA programs or NAS 411-1 for all other programs.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3 and table I).

b. Conformance inspection (see 4.4 and table I).

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TABLE I. First article and conformance inspections.

Characteristic	First Article	Conformance	Requirement	Verification
Workmanship	X	X	3.3.1	4.5.1.1
Size	X	X	3.3.2	4.5.1.2
Thickness	X	X	3.3.3	4.5.1.3
Symmetry	X	X	3.3.4	4.5.1.4
Flatness	X	X	3.3.5	4.5.1.5
Squareness	X	X	3.3.6	4.5.1.6
Areal Weight	X	X	3.3.7	4.5.1.7
Color	X	X	3.3.8	4.5.1.8
Dielectric Strength	X		3.4.1	4.5.2.1
Voltage	X		3.4.2	4.5.2.2
Combustibility	X		3.5	4.5.4
Flame Spread Index	X		3.5.1	4.5.3.2
Smoke Optical Density	X		3.5.2	4.5.3.3
Fire Gas Toxicity	X		3.5.3	4.5.3.4
Water Resistance	X		3.6.1	4.5.5.1
Moisture Absorption	X		3.6.2	4.5.5.2
Resistance to Cleaning Agents	X		3.6.3	4.5.5.3
Heat and Light Resistance	X		3.6.4	4.5.5.4
Light Reflectance	X		3.6.5	4.5.5.5
Impact Resistance	X		3.6.6	4.5.5.6
Shock Resistance	X		3.6.7	4.5.5.7
Slip Resistance	X		3.6.8	4.5.5.8
Wear Resistance	X		3.6.9	4.5.5.9
Roller Resistance	X		3.6.10	4.5.5.10
Short Beam Shear Strength	X		3.7.1	4.5.6.1
Long Beam Flexure Strength	X		3.7.2	4.5.6.2
Sandwich Compression Strength	X		3.7.3	4.5.6.3
Edgewise Compression Strength	X		3.7.4	4.5.6.4
Flatwise Tensile Strength	X	X	3.7.5	4.5.6.5
Maximum Panel Deflection	X		3.7.6	4.5.6.6
Panel Load Performance	X		3.7.7	4.5.6.7
Permanent Deflection	X		3.7.7.1	4.5.6.7
Machinability	X		3.8	4.5.7
Interface Capability	X		3.9	4.5.8
Toxicity and Prohibited Materials	X		3.10	4.5.9

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4.2 Inspection conditions. Samples for first article and conformance inspection and testing shall be conditioned at 73±2 °F, 50±5 percent relative humidity. Unless otherwise specified (see 6.2), testing shall be conducted under these same conditions.

4.2.1 Lot. A lot shall consist of all units of the same type, produced under similar conditions and ready for inspection or shipment at one time. When specified (see 6.2), the lot size shall be expressed in the number of false deck panels.

4.2.2 Sampling for inspection of panel. At a minimum, samples from each lot shall be randomly selected as specified in table II and inspected as specified in 4.5.

TABLE II. Sampling for inspection.

Lot Size [1/]	**Sample Size** [2/]
1 to 170	5
171 to 288	6
289 to 544	8
545 to 960	10
961 and above	12

NOTES:

1/ When the lot size is less than or equal to the sample size, 100-percent inspection is required.

2/ Refer to MIL-STD-1916 for increased sample size numbers if there is a need for tightened inspection due to non-conforming panels.

4.2.2.1 Destructive test sampling. In the case of the requirement for a destructive test to be performed as a conformance test of a panel, representative materials from the lot production may be used.

4.3 First article inspection. First article inspection shall be performed on the false deck panels when a first article sample is required (see 3.1). This inspection shall include the examinations and tests of table I and 3.3.1.

4.4 Conformance inspection. Conformance inspection shall include the examinations of table I and 3.3.1. A sampling of false deck panels (see 4.5.1) shall require conformance testing. When specified (see 6.2), certification of the tests shall be provided to the Technical Authority (see 3.6, 6.2, and 6.3).

4.5 Test procedures.

4.5.1 Physical characteristics.

4.5.1.1 Workmanship. Materials shall be visually inspected for defects that may adversely affect the performance or serviceability of the finished product (see 3.3.1).

4.5.1.2 Size. The false deck panels shall be measured for proper dimensions using a device capable of measuring to the nearest 1/32 inch (see 3.3.2 and 6.2).

4.5.1.3 Thickness. The false deck panels shall be measured for proper thickness using a device capable of measuring to the nearest 0.001 inch. The thickness of each panel shall be measured at a minimum of six points distributed over the area. The average of the six (or more) of the thickness readings shall be computed and considered the thickness of the false deck panels. No individual readings shall vary by greater than 5 percent of the specified thickness (see 3.3.3 and 6.2).

4.5.1.4 Symmetry. The false deck panels shall be visually inspected to ensure that they are symmetric along the horizontal centerline (see 3.3.4).

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4.5.1.5 Flatness. Linear flatness shall be tested in accordance with test procedures established by the manufacturer and approved by the Technical Authority (see 3.3.5).

4.5.1.6 Squareness. The perpendicularity of the false deck panels' edges shall be tested in accordance with ASTM F2055 (see 3.3.6).

4.5.1.7 Areal weight. The weight of the false deck panels shall be tested using a device capable of weighing to the nearest 0.01 pound (see 3.3.7).

4.5.1.8 Color. The false deck panels' color shall be tested in accordance with ASTM D2244 (see 3.3.8).

4.5.2 Electrical performance.

4.5.2.1 Dielectric strength. The dielectric strength of the false deck panels shall be tested in accordance with MIL-DTL-15562 (see 3.4.1). Specimens shall be tested to failure in air between 2-inch disk electrodes with edges rounded to a radius of ¼ inch. The voltage shall be applied at a rate in accordance with MIL-DTL-15562. The test specimen shall be a minimum of 1 square foot in area and there shall be a minimum of five measurements made with no two measurements being less than 4 inches apart.

4.5.2.2 Voltage. The voltage of the false deck panels shall be tested in accordance with MIL-DTL-15562. The voltage test shall be performed on the entire area of every panel of material offered for test except for the peripheral area within 2 inches of the edge. The material shall be tested between electrodes consisting of rectangular metal sheets, having smoothly rounded edges and corners, of any convenient length. The width shall be such that the arcing around the edges of the material will not occur. The tests shall be made progressively until the entire area of the material (the entire panel) has been covered, with 15,000 volts alternating current being applied for 1 minute at each position of the electrodes. The contact pressure of the electrodes shall be adequate to obtain good electrode contact with the test specimen (see 3.4.2).

4.5.3 Fire performance.

4.5.3.1 Fire testing provisions. All fire tests specified in this document shall be conducted by an independent testing laboratory that is accredited to ISO/IEC 17025 and is acceptable to the Technical Authority. Accreditation shall be obtained from a recognized accreditation body such as American Association for Laboratory Accreditation (A2LA) (www.A2LA.org) or International Code Council's International Accreditation Services (IAS) (www.IASonline.org). The scope of accreditation shall include specific flammability and fire tests. All other fire test provisions shall be as specified (see 6.2 and 6.9).

4.5.3.2 Flame spread index. The flame spread index of type I false deck panels shall be tested in accordance with ASTM E162 (see 3.5.1).

4.5.3.3 Smoke optical density. The smoke optical density of type I false deck panels shall be tested in accordance with ASTM E662 (flaming and non-flaming modes) (see 3.5.2).

4.5.3.4 Fire gas toxicity. The fire gas toxicity of type I false deck panels shall be tested in accordance with T9070-AK-DPC-10/078-1. Gas sampling and concentrations shall be measured in accordance with ASTM E800 when tested in accordance with ASTM E662 in both flaming and non-flaming modes (see 3.5.3).

4.5.4 Combustibility. Type II false deck panels shall be tested in accordance with ISO 1182 to determine that no materials are combustible. Class 1 wear surface materials that meet the requirements of MIL-STD-1623 are not required to be tested for combustibility. Group 1 electrical insulating materials that meet all the requirements of MIL-DTL-15562 are not required to be tested for combustibility.

4.5.5 Environmental durability.

4.5.5.1 Water resistance. The water resistance shall be determined when the panels are immersed in water for 24 hours in accordance with ASTM D570. Five 3 by 3-inch samples of false deck panels shall be tested (see 3.6.1).

4.5.5.2 Moisture absorption. Moisture absorption shall be determined when immersed in water for 24 hours in accordance with ASTM D570. Five 3 by 3-inch samples of false deck panels shall be tested (see 3.6.2).

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4.5.5.3 Resistance to cleaning agents. The false deck panels shall be subjected to the resistance to cleaning agents test in accordance with MIL-PRF-32170 (see 3.6.3).

4.5.5.4 Heat and light resistance. The heat and light resistance of false deck panels shall be tested in accordance with ASTM F1514 and ASTM F1515 (see 3.6.4).

4.5.5.5 Light reflectance. The light reflectance of the false deck panels shall be tested in accordance with ASTM D523 at the 60-degree geometry (see 3.6.5).

4.5.5.6 Impact resistance. The impact resistance of the false deck panels shall be tested in accordance with ASTM D7766, procedure C, to simulate a hammer drop test using a 1-inch diameter tip to impart an impact energy of 150 inch-pounds (see 3.6.6).

4.5.5.7 Shock resistance. The shock resistance of the false deck panels shall be tested in accordance with MIL-DTL-901 (see 3.6.7). The typical shipboard installation configuration shall be used for this testing (see 6.2). In the case of form 1 (unfinished) panel, shock design rules should be established that consider size, shape, and static loading conditions and a shock test program should be designed to validate these design rules by shock testing bounding cases. In the case of form 2, shock design rules should be established that consider static loading conditions on the flexible infrastructure deck track and panel, and a shock test program should be designed to validate these design rules by shock testing bounding cases.

4.5.5.8 Slip resistance. Slip resistance of the false deck panels shall be tested in accordance with ASTM D2047 (see 3.6.8).

4.5.5.9 Wear resistance. Wear resistance shall be tested in accordance with ASTM D4060 using a CS-17 wheel, 1,000 cycles, and 2.2-pound (1-kilogram) load (see 3.6.9).

4.5.5.10 Rolling load resistance (wheel A and B). Rolling load resistance tests shall be performed in accordance with the test procedures of CISCA 109 using a panel size of 24 inches by 48 inches for the form 1 (unfinished) panels, the standard form 2 (finished) size and a weight of 300 pounds. When specified (see 6.2), the false deck panels may be exempt from rolling load resistance testing.

4.5.5.10.1 Wheel A. A rolling load of 300 pounds shall be applied through a 3-inch diameter by $1\frac{13}{16}$-inch wide caster for 10 cycles over the same path without deflecting more than 0.04 inch and without permanent deformation in excess of 0.02 inch.

4.5.5.10.2 Wheel B. A rolling load of 300 pounds shall be applied through a hard rubber-surfaced wheel 6 inches in diameter by 2 inches wide for 10,000 cycles over the same path. Permanent set at conclusion of the test shall not exceed 0.010 inch.

4.5.6 Mechanical performance. Composite materials are typically orthotropic. Testing shall consider any non-isotropic nature of the false deck panels. For hexagonal honeycomb materials, the L (ribbon) and W (transverse) shear properties can be significantly different. If there is directionality to the false deck panels, they shall be tested in the weaker direction, typically the width direction.

4.5.6.1 Short beam shear strength. Short beam shear strength of the false deck panels shall be tested in accordance with ASTM C393. Five specimens shall be tested in the standard three-point bend configuration. If there is directionality to the false deck panels, the panel shall be tested in the weakest direction. The information within ASTM C393 shall be used to ensure that the sample size and span distance are selected to ensure a core shear failure based on the specific geometry of the false deck panels (see 3.7.1).

4.5.6.2 Long beam sandwich flexure. Long beam sandwich flexure of the false deck panels shall be tested in accordance with ASTM D7249. Five specimens, 3 inches by 24 inches each, shall be tested in the standard four-point (quarter-span) bend configuration. If there is directionality to the false deck panels, it shall be tested in the width direction. The information within ASTM D7249 shall be used to ensure that the sample size and span distance are selected to ensure the desired face sheet failure mode (see 3.7.2).

4.5.6.3 Sandwich compression strength. Sandwich compression of the false deck panel shall be tested in accordance with ASTM C365. Five specimens, 2 inches by 2 inches each, shall be tested (see 3.7.3).

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4.5.6.4 Edgewise compression strength. Edgewise compression strength shall be tested in accordance with ASTM C364. Five specimens, 2 inches by 2 inches each, shall be tested (see 3.7.4).

4.5.6.5 Flatwise tensile strength. Flatwise tensile strength shall be tested in accordance with ASTM C297. Five specimens, 2 inches by 2 inches each, shall be tested (see 3.7.5).

4.5.6.6 Maximum panel deflection. The maximum panel deflection shall be determined by measuring the deflection at the center of the panel under test.

a. Form 1 (unfinished) – A 24- by 48-inch panel shall be supported at a width of 0.5-inch along all edges and installed according to the typical shipboard configuration with a 300-pound load applied uniformly over a 10-by 10-inch area in the center of the panel (see 3.7.6).

b. Form 2 (finished) – A typical false deck tile (10.375 inches by 24 inches) shall be supported at a width of 1-inch along the 24-inches edge of the tile and installed according to typical shipboard configuration with a 300-pound load applied uniformly over a 4-by 10-inch area in the center of a tile. The 10-inch length being parallel to the 24-inch support edge (see 3.7.6).

4.5.6.7 Panel load performance. The panel load performance testing shall be performed by using the same loading configuration as in 4.5.5.6 (see 3.7.7 and 3.7.7.1).

4.5.7 Machinability. The machinability of the false deck panel material shall be evaluated by performing a visual inspection of the cut surfaces after using standard hand-tools, such as circular saws and drills, to cut the panels to fit the shipboard configuration. Typical equipment that should be evaluated is listed in table III. The false deck panels shall be visually inspected to ensure that no signs of chipping, cracking, splintering, blistering, or delamination have occurred after standard machining (see 3.8).

TABLE III. Equipment to be considered for machinability.

Equipment Type	Details
Circular Saw	SSC model 5860-72:78 with 8" blade or Dewalt 7¼" Ripping Circular Saw Blade (36 number of teeth) or equivalent
Band Saw	No specific call-out
Saber/Jig Saw	Milwaukee Brand Jig Saw with carbide blades or equivalent
Drill Press	Jet drill press, 12-speed with 1 1/16-countersink drill bit, Walker drill press with countersink ⅜ drill bit, or equivalent
Hand Drill	Makita 18-volt Cordless or equivalent
Router	Porter Cable or equivalent
Sanding	Files, 3-inch grinders, hand sanding blocks

4.5.8 Interface capability. The interface capability of the false deck panels shall be determined by performing the shipboard installation set-ups required to support testing in 4.5.5.10, 4.5.6.6, and 4.5.6.7. Visual inspection shall be performed on the installations to determine that the false deck panels can support the shipboard integration system. Form 1 (unfinished) and form 2 (finished) false deck panels shall be tested in their respective shipboard configurations (see 3.9).

4.5.9 Toxicity and prohibited materials.

4.5.9.1 Toxicity. A Health Hazard Assessment (HHA) will be conducted to ensure conformance to 3.10 as specified (see 6.2). The Navy and Marine Corps Public Health Center (NMCPHC) will evaluate the false deck panels using data provided by the manufacturer/distributor to the NMCPHC (see 3.10 and 6.7).

4.5.9.2 Prohibited materials. Prohibited materials shall be verified as specified (see 3.10 and 6.2).

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The false deck panels covered by this document are intended for use as a component of the false deck system that performs numerous functions in Navy shipboard environments. The false deck panels interface with the false deck support system to provide a level walking surface free of trip and slip hazards to ensure safety of personnel. The false deck panels meet Navy fire performance requirements and provide electrical safety in accordance with S9086-VG-STM-010/634, Naval Ships' Technical Manual (NSTM) Chapter 634, Deck Coverings, General and S9086-KC-STM-010/300, NSTM Chapter 300, Electrical Plant, General when utilized in electronics spaces and allow cable penetrations to support power connections to electrical equipment. This specification supersedes any reference to NAVSEA Drawing 803-6397418, Nomex Honeycomb False Deck Panel Requirements. False deck panels can be classified into types, classes, forms, groups and areal weights. The type is related to the level of combustibility of the materials used in the false deck panels. Type I is fabricated from any combustible material, and type II is manufactured using only non-combustible materials in accordance with ISO 1182. The class is related to whether the wear surface on the panel is in accordance with MIL-STD-1623. A class 1 material will meet the fire performance requirements for interior finish material and furnishings whereas class 2 does not. The form of the false deck panels is defined as a finished or unfinished panel. The form 1 is unfinished and can be cut to size with whereas the form 2 is a smaller panel with edges that are fabricated to fit into the flexible infrastructure system. The group of the false deck panel defines whether the panels are to be installed in electrical or non-electrical spaces. Group 1 provides safety inspection from accidental exposure to electrical potential in electrical spaces whereas group 2 panels are considered general-purposed for installation in non-electrical spaces. The areal weight of the panels comes in three levels from 1 to 3. Areal weight 1 is the lightest deck panels and is where weight is most critical, areal weight 2 is where weight is less critical, and areal weight 3 is where weight is not critical. The areal weight levels are defined by specific areal weight density values.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Classification (type, class, form, group, and weight) (see 1.2).

c. When first article inspection is required (see 3.1).

d. Panel size, symmetry, and color (see 3.3.2, 3.3.3, 3.3.4, and 3.3.8).

(1) If the acquisition document allows a non-symmetric false deck panels, such as a wear surface only on one side, the Technical Authority and the procuring activity can decide to allow a non-symmetric false deck panel if it can be proved to be technically acceptable.

(2) If the procuring activity desires that a particular pattern be matched, then a sample pattern should be provided to the manufacturer. The manufacturer should be requested to provide a sample of the pattern to be produced subject to approval by the procuring activity. In determining acceptance of the manufacturer's sample, the procuring activity should recognize that an exact match is not necessary and that there will be variations in colors, mottling, and marbleization that should be considered to be acceptable.

e. Requirements for an HHA (see 3.10 and 4.5.9).

f. Requirement for prohibited materials verification (see 3.10 and 4.5.9).

g. Inspection conditions, if other than specified herein, or as specified by the specific test specification (see 4.2).

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h. Lot size (see 4.2.1).

i. Sampling for conformance (see 4.2.2).

j. When certification of the tests is required (see 4.4).

k. Size requirements for inspection (see 4.5.1.2).

l. Thickness requirements for inspection (see 4.5.1.3).

m. Additional fire testing provisions (see 4.5.3.1 and 6.9).

n. The intended shipboard installation configurations to be used as the bounding cases for the shock testing (see 4.5.5.7).

o. Exemption from rolling load resistance testing (see 4.5.5.10).

p. Toxicity requirements (see 4.5.9).

q. Packaging requirements (see 5.1).

r. Reporting requirements (see 6.4).

s. Safety Data Sheet (SDS), when required (see 6.6).

6.3 Technical Authority. The Technical Authority has the authority, responsibility, and accountability to establish, monitor, and approve technical standards, tools, and processes in conformance to higher authority policy, requirements, architectures, and standards. NAVSEA 05S may be contacted to determine the appropriate Technical Authority for any given requirement related to equipment and systems of U.S. Navy ships and submarines.

6.4 Reporting of test results. Each test report should include a complete description of the materials tested. Panels should be described in detail as to the type of material configuration, orientation, and density (see 6.2). Descriptions of sandwich constructions should include the type, size, and orientation of the facings, the procedure and materials used to bond the facings to the core, and detailed description of the core and its orientation. If any additional materials or steps are needed to meet the requirements in this specification after the false deck panels are modified by machining, then this should be explicitly cited in the test report.

6.5 Changes to product. Any change in basic ingredients or processes which would affect compliance with this specification should be reported to both the contracting activity and the Technical Authority and approved prior to production.

6.6 SDS. When required (see 6.2), contracting officers will identify those activities requiring a company's completed SDS prepared in accordance with FED-STD-313. In order to obtain the SDS, FAR Clause 52.223-3 must be in the contract.

6.7 Toxicity evaluation. The NMCPHC requires sufficient information to permit an HHA of the product. Upon completion of the HHA, a copy will be provided by the NMCPHC to the Government for evaluation. The HHA process is described on the NMCPHC's website, http://www.med.navy.mil/sites/nmcphc/industrial-hygiene/Pages/health-hazard-assessment.aspx.

6.8. Fire performance requirements (see 3.5). Deviations from fire gas toxicity requirements and maximum flame spread may be requested from the Technical Authority when appropriate. Where the values achieved in 4.5.3.4 are only slightly out of range of the values specified in 3.5.3, the test results and the particulars of the materials may be presented to NAVSEA 05P for consideration of the acceptability of the results. If a Nomex core, $\frac{3}{16}$, 0.5 lbs/ft^2 is used, the flame spread requirement is 45 (maximum). If phenolic infused balsa core is used, the flame spread requirement is 45 (maximum) for the core with measurements taken from both the top and the bottom to ensure complete infusion of the core.

6.9 Additional fire testing provisions. The Technical Authority reserves the right to witness the tests and perform any of the tests set forth herein where such testing is deemed necessary to assure compliance to prescribed requirements. A2LA (www.A2LA.org) or International Code Council's IAS (www.IASonline.org) maintain a full list of the approved laboratories that are accredited to ISO/IEC 17025 and provide details on how to obtain and maintain accreditation (see 6.2).

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6.10 Subject term (key word) listing.

Access panel
Deck tile
Dielectric
False deck
Fire performance
Flexible infrastructure

CONCLUDING MATERIAL

Custodians:
Army – AV
Navy – SH

Review activities:
Army – MI
DLA – CQ

Civil agency:
GSA – FAS

Preparing activity:
Navy − SH
(Project 5670-2020-001)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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